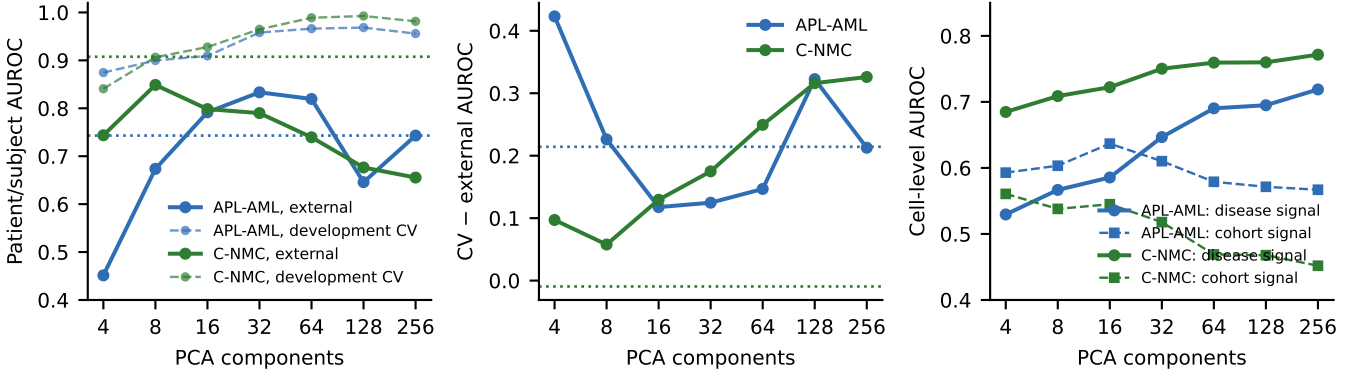
(a) Development vs. external
Patient/subject AUROC
1.0
0.9
0.8
0.7
0.6
0.5
0.4
APL-AML, external
APL-AML, development CV
C-NMC, external
C-NMC, development CV
4
8
16
32
64
128
256
PCA components
(b) Generalization gap
CV − external AUROC
0.4
0.3
0.2
0.1
0.0
APL-AML
C-NMC
4
8
16
32
64
128
256
PCA components
(c) What the components keep
Cell-level AUROC
0.8
0.7
0.6
0.5
0.4
APL-AML: disease signal
APL-AML: cohort signal
C-NMC: disease signal
C-NMC: cohort signal
4
8
16
32
64
128
256
PCA components